



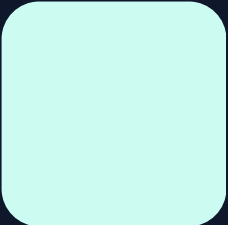
DAY 4



Model Integration & Error Analysis

Garbage Classification — From Raw Image to Prediction and Error Investigation

Week 8 • AI & ML Internship



Prepared By Sadeel Ardah



Day 3 prepares the image. Day 4 connects everything.

Day 3 — Preprocessing

- Read image
- Resize to 224×224
- BGR → RGB
- Normalize pixels



Day 4 — Integration & Analysis

- Build one predict() pipeline
- Keep preprocessing consistent
- Analyze errors
- Inspect wrong predictions

Main idea: raw input should enter one side and a prediction should come out the other.

From scattered notebook cells to one coherent pipeline

Before Integration

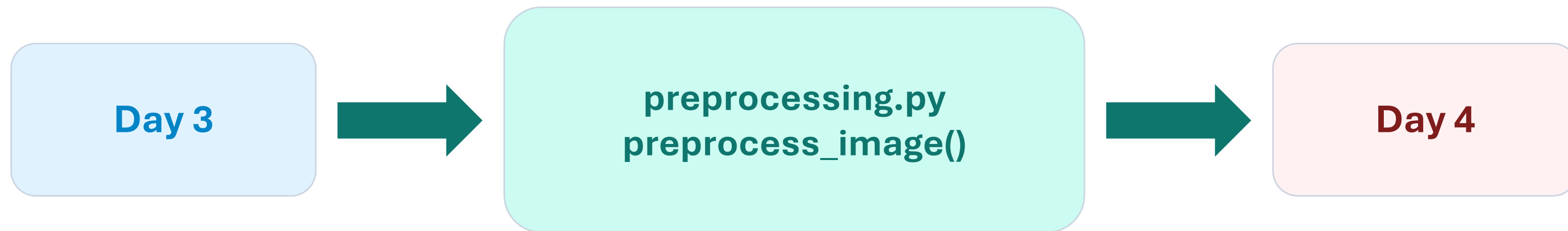
Cell 1 → preprocessing
Cell 2 → feature preparation
Cell 3 → model
Cell 4 → prediction
Cell 5 → evaluation



After Integration

Raw Input
↓
One ordered pipeline
↓
Prediction

One source of truth for preprocessing

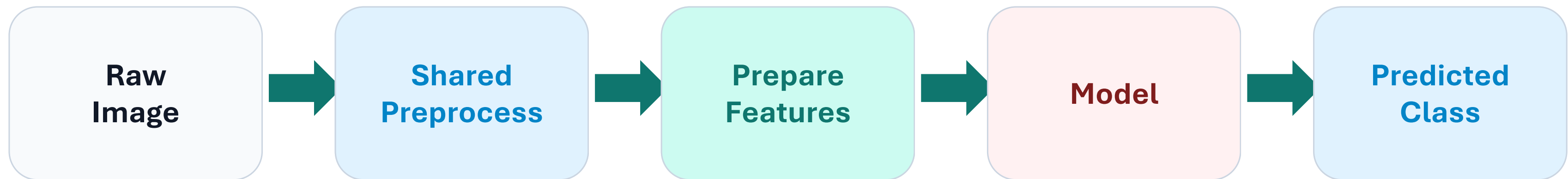


Why this matters

Resize, color conversion, and normalization are defined once and reused everywhere.

Same preprocessing code = lower risk of silent mismatch.

One raw garbage image → one prediction



```
def predict(raw_input):  
    processed = preprocess(raw_input)  
    prepared  = prepare(processed)  
    return model.predict(prepared)
```


The same model can fail if preprocessing changes

Training — Correct

Resize → RGB → Normalize



Model

Prediction — Wrong

Different size → BGR → No normalization



Model

TRAINING / SERVING SKEW

Golden rule: prediction-time preprocessing must match training-time preprocessing.

Accuracy tells us how often. Error analysis tells us where and why.



Question 1

Which classes are confused?

Question 2

Is one class weaker than others?

Question 3

Data issue or model weakness?

Actual class vs. predicted class

		Predicted →			
		Paper	Cardboard	Plastic	Metal
Actual ↓	Paper	18	4	1	0
	Cardboard	5	16	1	0
	Plastic	1	0	20	2
	Metal	0	0	3	19

What to look for

Large off-diagonal values reveal repeated confusion between classes.

Illustrative example

Cardboard → Paper appears often, so that pair deserves inspection.

Count the repeated actual → predicted mistakes

Error Pair	Count	What it suggests
Cardboard → Paper	5 examples	The two classes may share similar color or texture.



Now inspect the images themselves.

The images explain what the matrix cannot

Example A

Actual: Cardboard
Predicted: Paper

Question:
Are texture and color genuinely
similar?

Example B

Actual: Green Glass
Predicted: White Glass

Question:
Did lighting change the
apparent color?

Example C

Actual: Plastic
Predicted: Metal

Question:
Is the object reflective or
unusually cropped?

Qualitative inspection turns a wrong prediction into an actionable explanation.

Not every wrong prediction has the same cause

Data Issue

- Mislabeled image
- Bad crop
- Very dark / blurry
- Ambiguous object
- Unusual background

Model Weakness

- Image is clear
- Label is correct
- Class patterns are visible
- Model still confuses it

This decision guides the fix: improve the data or improve the model.

What I Learned Today

- 1 Model integration
- 2 End-to-end predict() pipeline
- 3 Reuse of Day 3 preprocessing
- 4 Training / serving consistency
- 5 Training / serving skew
- 6 Error analysis beyond accuracy
- 7 Confusion matrix interpretation
- 8 Misclassified-example inspection
- 9 Data issue vs. model weakness

KEY TAKEAWAY

A professional ML workflow keeps preprocessing consistent and studies mistakes instead of stopping at accuracy.